



TechBiz Academy

Week 3 · Task 9

Vulnerability Scanning Basics

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1. Introduction

Vulnerability assessment is the process of identifying security weaknesses in a system before attackers can exploit them. In this lab, Nmap was used to scan a vulnerable virtual machine (Metasploitable2), identify security issues, and research publicly known Common Vulnerabilities and Exposures (CVEs). Additionally, Nessus Essentials was installed as an optional vulnerability scanning tool.

2. Objective

The objectives of this assessment were to:

- Perform a vulnerability scan using Nmap.
 - Identify potential vulnerabilities on the target system.
 - Research the discovered vulnerabilities using their CVE numbers.
 - Install Nessus Essentials (optional).
 - Understand the importance of vulnerability management.
-

3. Environment Setup

Component	Details
Scanner Machine	Kali Linux
Target Machine	Metasploitable2
Target IP	192.168.1.6

Component	Details
Tool Used	Nmap 7.98
Optional Tool	Nessus Essentials

4. Vulnerability Scan using Nmap

Command Used

```
nmap -sV --script vuln -T4 192.168.1.6
```

Screenshot

```
Nmap done: 1 IP address (1 host up) scanned in 1.12 seconds
(hamza@hamza)-[~]
$ nmap -sV --script vuln -T4 192.168.1.6
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-22 08:34 -0400
Nmap scan report for 192.168.1.6
Host is up (0.00038s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-vsftpd-backdoor:
| VULNERABLE:
|   vsFTPD version 2.3.4 backdoor
|   State: VULNERABLE (Exploitable)
|   IDs: CVE:CVE-2011-2523 BID:48539
|   vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|   Disclosure date: 2011-07-03
|   Exploit results:
|     Shell command: id
|     Results: uid=0(root) gid=0(root)
|   References:
|     https://www.securityfocus.com/bid/48539
|     http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|     https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|_ vulners:
|   vsftpd 2.3.4:
|     PACKETSTORM:162145 10.0 https://vulners.com/packetstorm/PACKETSTORM:162145 *EXPLOIT*
|     FEA24DDE-72FF-5445-B198-9E2D5322DC86 10.0 https://vulners.com/githubexploit/FEA24DDE-72FF-5445-B198-9E2D5322DC86 *EXPLOIT*
|     EDB-ID:49757 10.0 https://vulners.com/exploitdb/EDB-ID:49757 *EXPLOIT*
|     ED5791E3-8935-5632-A403-CB7E7FA26CB0 10.0 https://vulners.com/githubexploit/ED5791E3-8935-5632-A403-CB7E7FA26CB0 *EXPLOIT*
|     EC73033B-7726-59C3-828B-D82BBB955DB9 10.0 https://vulners.com/githubexploit/EC73033B-7726-59C3-828B-D82BBB955DB9 *EXPLOIT*
|     E9B0AEBB-5138-50BF-8922-2D87E3C046DD 10.0 https://vulners.com/githubexploit/E9B0AEBB-5138-50BF-8922-2D87E3C046DD *EXPLOIT*
|     E356CCEC-7C81-5A77-8CE4-87DF4DB21897 10.0 https://vulners.com/githubexploit/E356CCEC-7C81-5A77-8CE4-87DF4DB21897 *EXPLOIT*
|     DC8CF54B-5E0D-55F4-B5C1-920406A0FF9F 10.0 https://vulners.com/githubexploit/DC8CF54B-5E0D-55F4-B5C1-920406A0FF9F *EXPLOIT*
|     D0DBA05F-750F-5F9B-9C69-65E8FA6977D5 10.0 https://vulners.com/cve/CVE-2011-2523 *EXPLOIT*
```

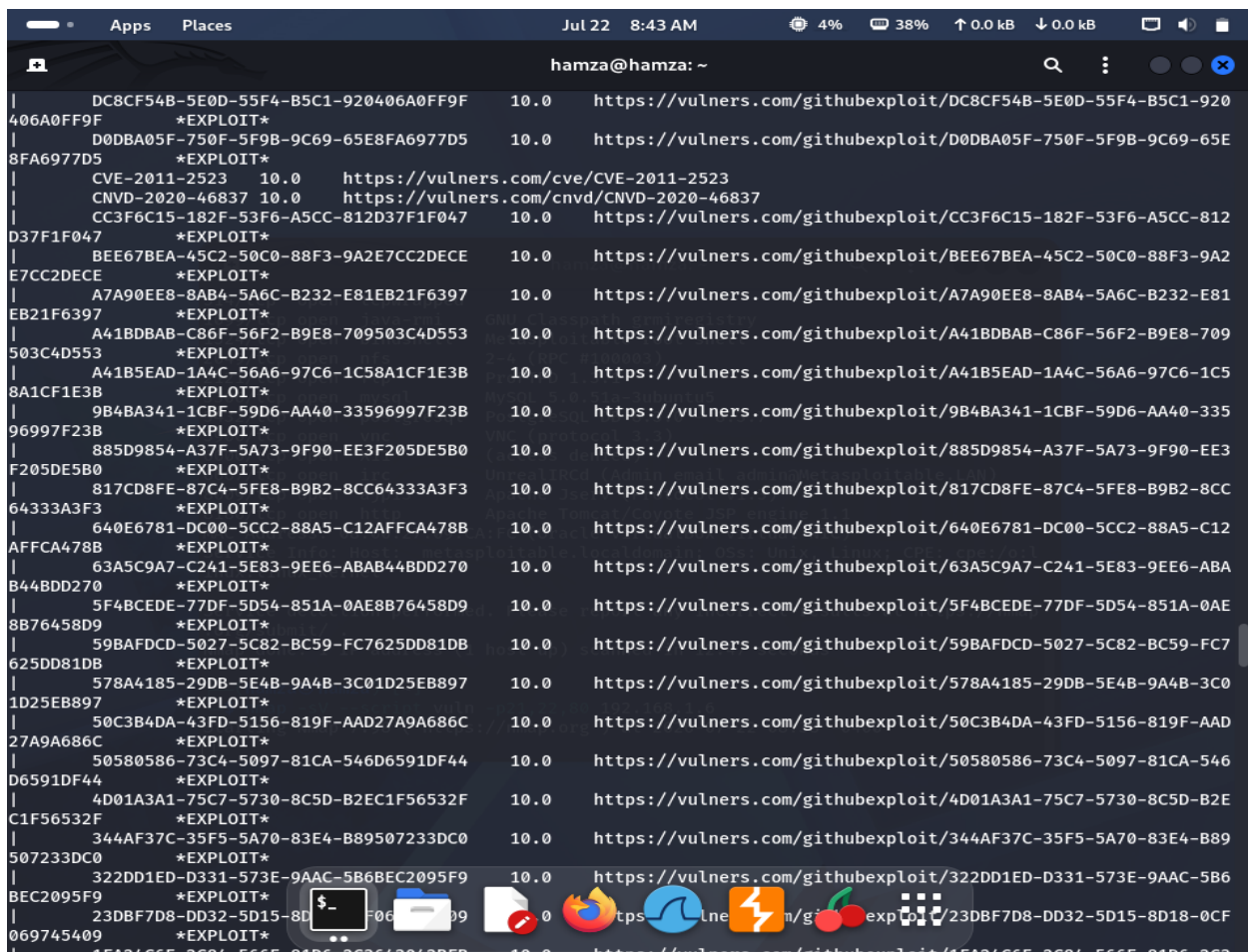
Caption: Figure 1 – Nmap command execution and initial vulnerability scan output.

After the scan completed, Nmap identified multiple open ports and vulnerable services running on the target machine.

The major open services included:

Port	Service	Version
21	FTP	vsFTPD 2.3.4
22	SSH	OpenSSH 4.7p1
23	Telnet	Open
25	SMTP	Open
53	DNS	ISC BIND 9.4.2
80	HTTP	Apache HTTP Server 2.2.8

Screenshot



Caption: Figure 2 – Detailed Nmap vulnerability scan results.

5. Identified Vulnerabilities

Vulnerability 1

Service

FTP (vsFTPD 2.3.4)

CVE

CVE-2011-2523

Description

This version of vsFTPD contains a malicious backdoor that allows unauthorized remote shell access to the target system.

Severity

Critical

Recommendation

- Upgrade vsFTPD.
 - Disable FTP if unnecessary.
 - Use SFTP instead.
-

Vulnerability 2

Service

SSL 3.0

CVE

CVE-2014-3566 (POODLE)

Description

The server supports SSL 3.0, making encrypted communications vulnerable to the POODLE attack.

Severity

High

Recommendation

- Disable SSL 3.0.
 - Enable TLS 1.2 or TLS 1.3.
-

Vulnerability 3

Service

Apache HTTP Server 2.2.8

CVE

CVE-2011-3192

Description

Apache 2.2.8 is outdated and vulnerable to multiple attacks, including the Apache Range Header Denial-of-Service vulnerability.

Severity

High

Recommendation

- Upgrade Apache to a supported version.
 - Apply security patches.
 - Disable unnecessary HTTP methods.
-

6. CVE Research

Vulnerability	CVE Number	Summary
vsFTPD Backdoor	CVE-2011-2523	Allows unauthorized remote shell access through a backdoor.
SSL POODLE	CVE-2014-3566	Allows decryption of encrypted SSL 3.0 communications.
Apache HTTP Server	CVE-2011-3192	Denial-of-Service vulnerability affecting Apache 2.2.x.

7. Nessus Essentials Installation (Optional)

The Nessus Essentials vulnerability scanner was installed successfully on Kali Linux.

Installation Command

```
sudo dpkg -i Nessus-10.12.1-debian10_amd64.deb
```

```
hamza@hamza: ~/Downloads
Nessus-10.12.1-debian10_amd64.deb

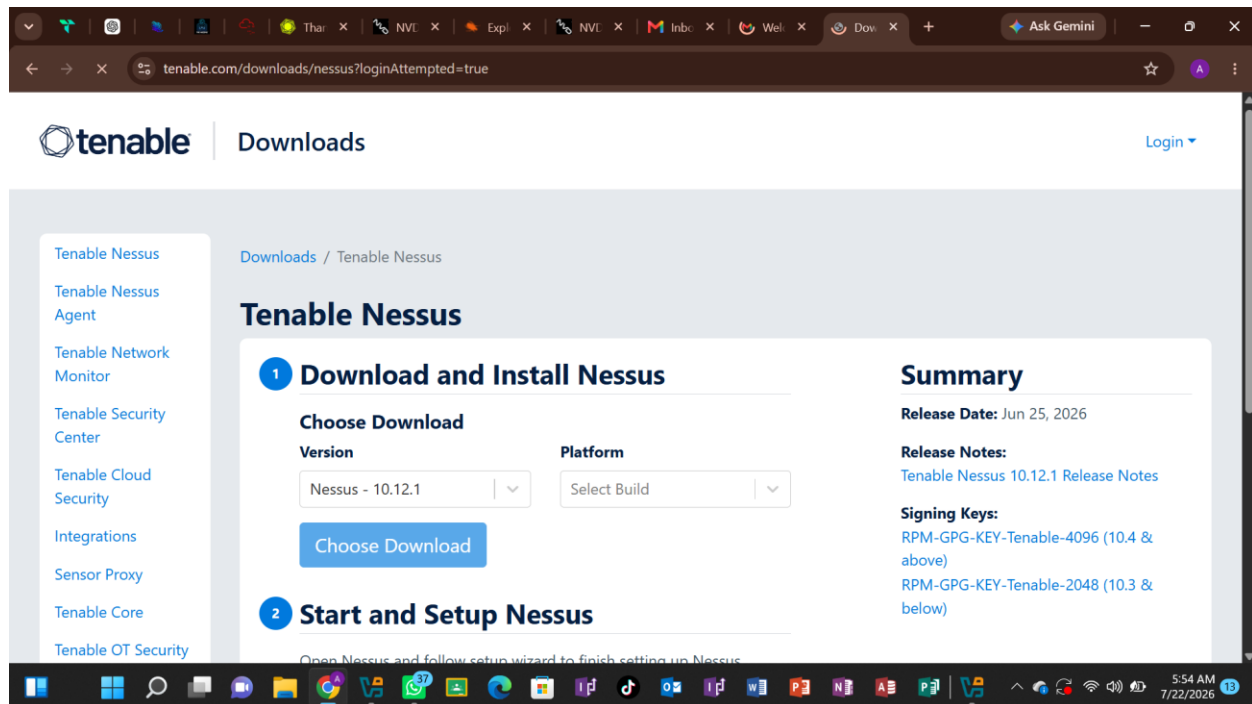
(hamza@hamza) - [~/Downloads]
$ sudo dpkg -i Nessus-*.deb
Selecting previously unselected package nessus.
(Reading database... 592219 files and directories currently installed.)
Preparing to unpack Nessus-10.12.1-debian10_amd64.deb...
Unpacking nessus (10.12.1)...
Setting up nessus (10.12.1)...
HMAC : (KAT_Integrity) : Pass
HMAC : (Module_Integrity) : Pass
SHA1 : (KAT_Digest) : Pass
SHA2 : (KAT_Digest) : Pass
SHA3 : (KAT_Digest) : Pass
AES_GCM : (KAT_Cipher) : Pass
AES_ECB_Decrypt : (KAT_Cipher) : Pass
RNG : (Continuous_RNG_Test) : Pass
RSA : (KAT_Signature) : Pass
ECDSA : (KAT_Signature) : Pass
ECDSA : (KAT_Signature) : Pass
DSA : (KAT_Signature) : Pass
TLS13_KDF_EXTRACT : (KAT_KDF) : Pass
TLS13_KDF_EXPAND : (KAT_KDF) : Pass
TLS12_PRF : (KAT_KDF) : Pass
PBKDF2 : (KAT_KDF) : Pass
SSHKDF : (KAT_KDF) : Pass
KBKDF : (KAT_KDF) : Pass
HKDF : (KAT_KDF) : Pass
SSKDF : (KAT_KDF) : Pass
X963KDF : (KAT_KDF) : Pass
X942KDF : (KAT_KDF) : Pass
HASH : (DRBG) : Pass
CTR : (DRBG) : Pass
HMAC : (DRBG) : Pass
DH : (KAT_KA) : Pass
ECDH : (KAT_KA) : Pass
RSA_Encrypt : (KAT_AsymmetricCipher) : Pass
RSA_Decrypt : (KAT_AsymmetricCipher) : Pass
RSA_Decrypt : (KAT_AsymmetricCipher) : Pass
name: OpenSSL FIPS Provider
version: 3.1.2
build: 3.1.2
INSTALL PASSED
Unpacking Nessus Scanner Core Components...
```

Caption: Figure 3 – Installing Nessus Essentials on Kali Linux.

Start Nessus Service

```
sudo systemctl start nessusd
```


Screenshot



Caption: Figure 4 – Nessus service setup or activation page.

Open Nessus Web Interface

`https://localhost:8834`

Complete the activation process using a Nessus Essentials activation code, then create a **Basic Network Scan** targeting the Metasploitable2 IP address.

8. Conclusion

The vulnerability assessment successfully identified several known security weaknesses on the Metasploitable2 system using Nmap. The scan revealed critical vulnerabilities such as the **vsFTPD 2.3.4 Backdoor (CVE-2011-2523)**, **SSL POODLE (CVE-2014-3566)**, and an outdated **Apache HTTP Server (CVE-2011-3192)**. These findings demonstrate the importance of keeping software updated, disabling insecure protocols, and regularly performing vulnerability assessments. The optional installation of Nessus Essentials further demonstrated how professional vulnerability scanners can be used to automate detection and reporting of security issues.

9. Learning Resources & References

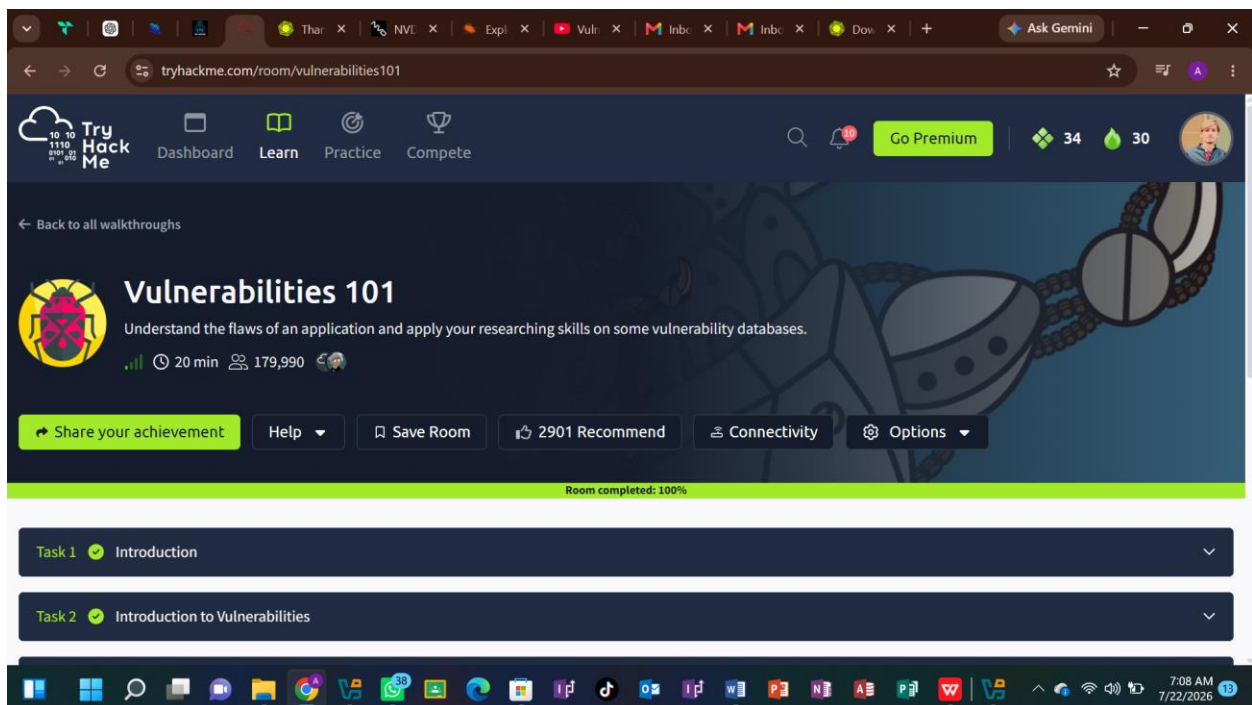
During this vulnerability assessment, the following educational resources and documentation were used to understand vulnerability scanning, CVEs, and the use of Nmap and Nessus.

Resource 1 – TryHackMe: Vulnerabilities 101

Purpose

This free TryHackMe room provides an introduction to common vulnerabilities, CVEs, vulnerability assessment, and security best practices.

Screenshot



Caption: Figure 5 – TryHackMe: Vulnerabilities 101.

Resource 2 – Nmap NSE (Nmap Scripting Engine) Reference

Purpose

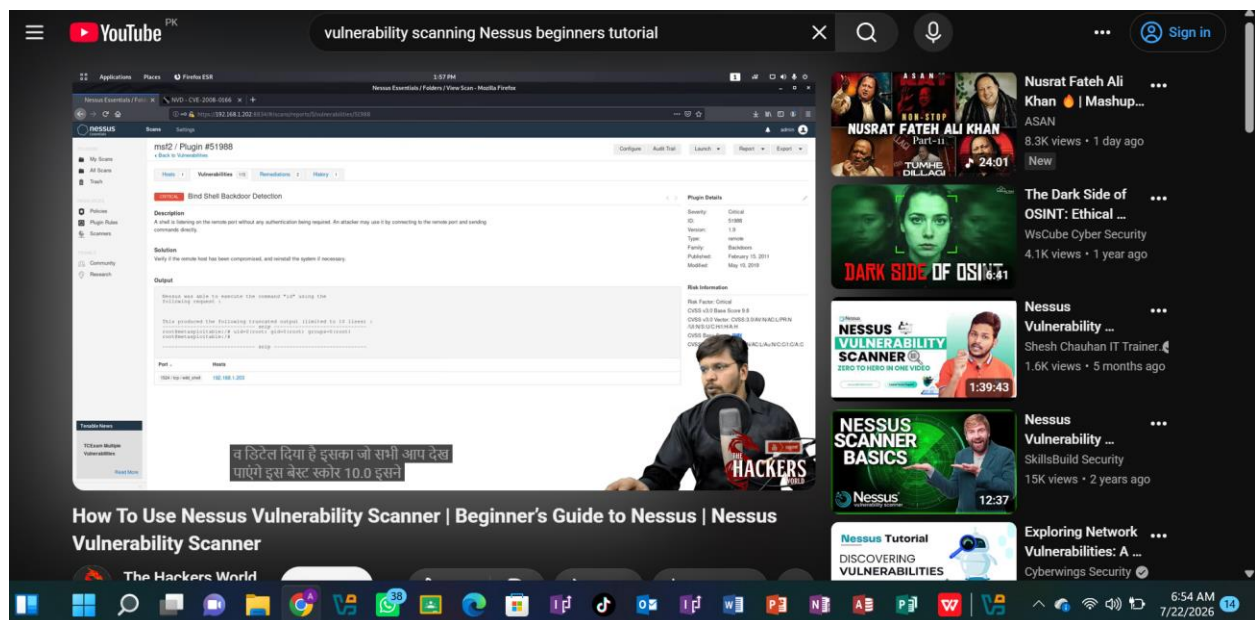
The official Nmap NSE documentation was used to understand how vulnerability scripts (`--script vuln`) work and how they identify known security issues.

Resource 3 – YouTube Tutorial: Vulnerability Scanning with Nessus (Beginner Tutorial)

Purpose

This tutorial demonstrated the installation, configuration, and execution of vulnerability scans using Nessus Essentials.

Screenshot



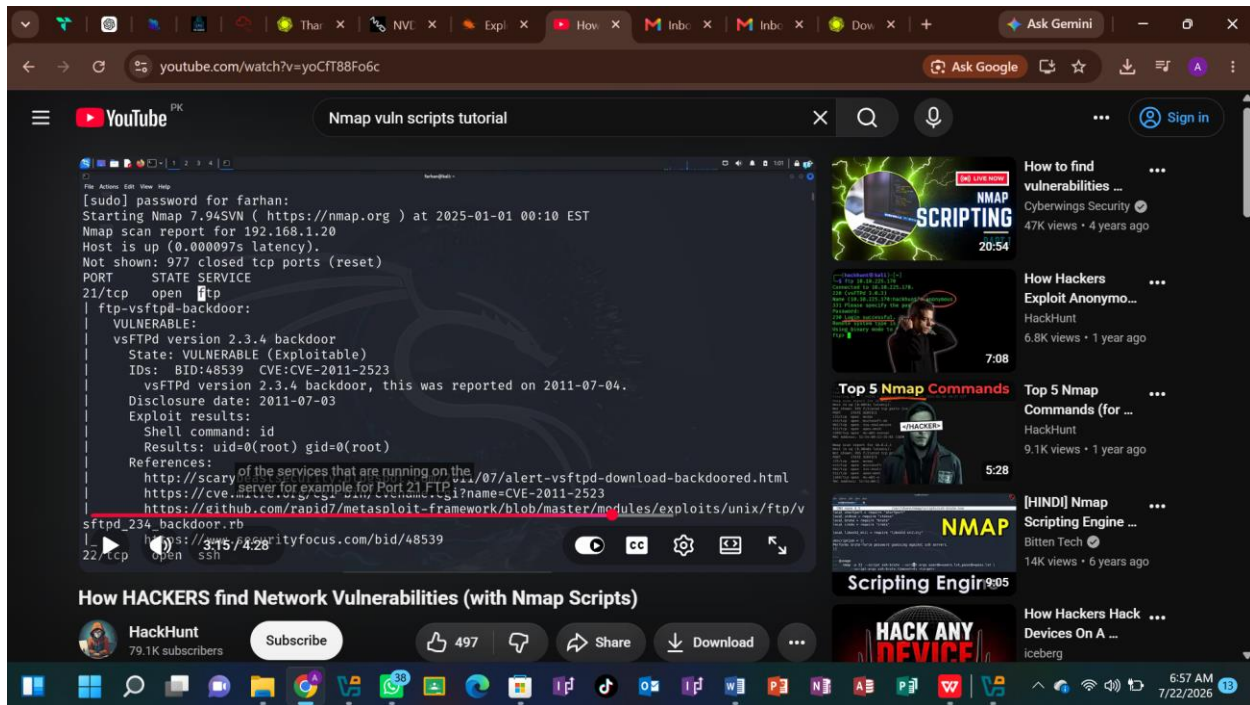
Caption: Figure 7 – Nessus Beginners Tutorial.

Resource 4 – YouTube Tutorial: Nmap Vulnerability Scripts

Purpose

This tutorial explained how to use Nmap vulnerability scripts to detect known vulnerabilities and interpret scan results.

Screenshot



Caption: Figure 8 – Nmap Vulnerability Scripts Tutorial.

10. References

1. TryHackMe. *Vulnerabilities 101*.
2. Nmap Project. *Nmap Scripting Engine (NSE) Documentation*.
3. Tenable. *Nessus Essentials Documentation*.
4. MITRE. *Common Vulnerabilities and Exposures (CVE)*.
5. National Vulnerability Database (NVD). *CVE Details and Security Advisories*.
6. YouTube. *Vulnerability Scanning with Nessus – Beginners Tutorial*.
7. YouTube. *Nmap Vulnerability Scripts Tutorial*.